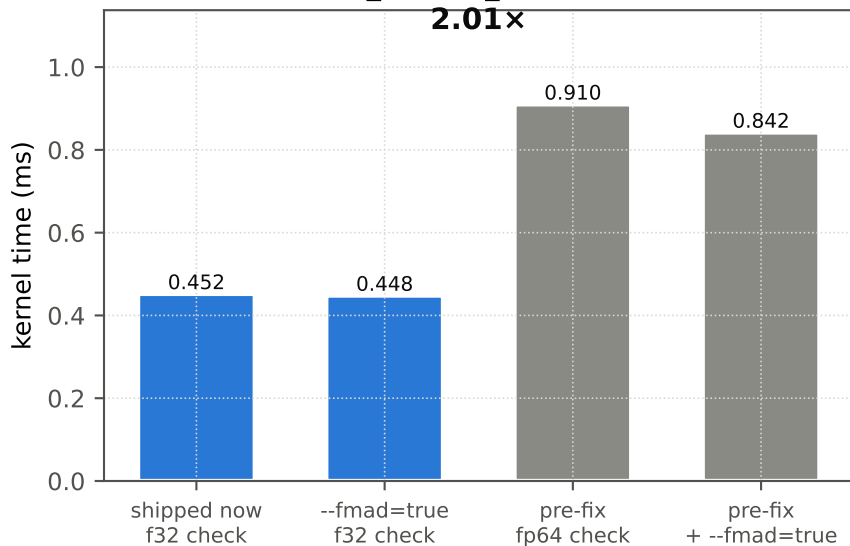
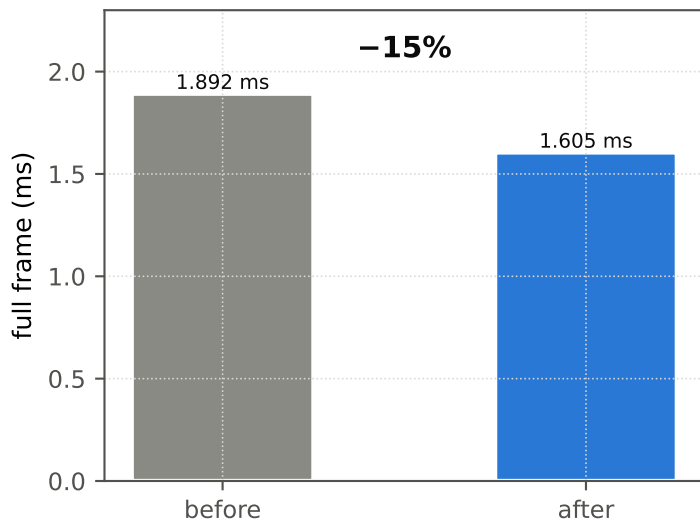


fractal_kernel_f32 ablation



Effect on the whole frame



mandelbrot_f32 called the fp64 in_period3_bulb once per pixel. On Ampere GeForce, which runs fp64 at 1/64 the fp32 rate, those 8 operations cost as much as the entire 1000-iteration f32 loop. Replacing them with an f32 predicate is bit-identical — zero differing pixels of 2,073,600 across seven viewports including three centred on the period-3 boundary arc. The full-frame gain is only 15% because the frame is readback-bound (Figure 3).